



Department of Information Technology

Academic Year: 2025- 2026(Odd Semester)

Semester, Degree & Branch: V Semester, B.Tech & IT

Course Code & Title: CS3551 & Distributed Computing

Name of the Faculty member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Synchronous versus Asynchronous Executions
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** One Minute Paper
- **Justification:**
 - The One-Minute Paper activity provides students with a brief, focused opportunity to summarize and reflect on what they have learned during the session, helping reinforce key concepts(making those crucial ideas stronger and more memorable).
 - Synchronous versus Asynchronous Executions is a fundamental topic that helps students understand different approaches to task execution and timing in computing systems.
 - This activity was chosen as a quick and effective recap tool to reinforce foundational knowledge and assess students understanding of synchronous and asynchronous execution models.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The faculty introduced the topic by explaining the concepts of synchronous and asynchronous executions, including their definitions, differences, use cases, and significance in computing systems.
 - After the explanation, students were asked to summarize the topic within 60 seconds on paper.
 - Sample responses by Mr. Gopi S and Ms. Divya Sri J are shown in Figure 1.
 - The faculty reviewed the responses. Students who wrote incorrect or unclear summaries were guided and asked to rewrite them for better understanding.

• CO – PO / PSO mapping (specific to this activity):

CO	PO1	PO2	PO3	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	2	1	3	3	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO10	PO12	PSO1	PSO2	PSO3
	3	3	1	2	1	3	3	2
Justification for correlation	Students apply fundamental engineering knowledge to solve problems involving synchronous and asynchronous execution models in computing systems.	They analyze complex timing and execution issues by leveraging synchronous and asynchronous techniques.	Students provide solutions to problems by understanding global states, synchronization, and asynchronous behaviors in distributed environments.	They explore and deepen technical knowledge of synchronous and asynchronous execution models and their implications in distributed computing.	Students understand the historical development and evolution of execution models and their role in modern parallel and distributed systems.	They apply synchronous and asynchronous concepts in developing mobile and distributed applications.	Students deliver reliable solutions addressing timing and concurrency issues in IT systems using knowledge of execution models.	They demonstrate proficiency in designing systems that effectively handle data processing using synchronous and asynchronous architectures.

• Images / Screenshot of the practice:

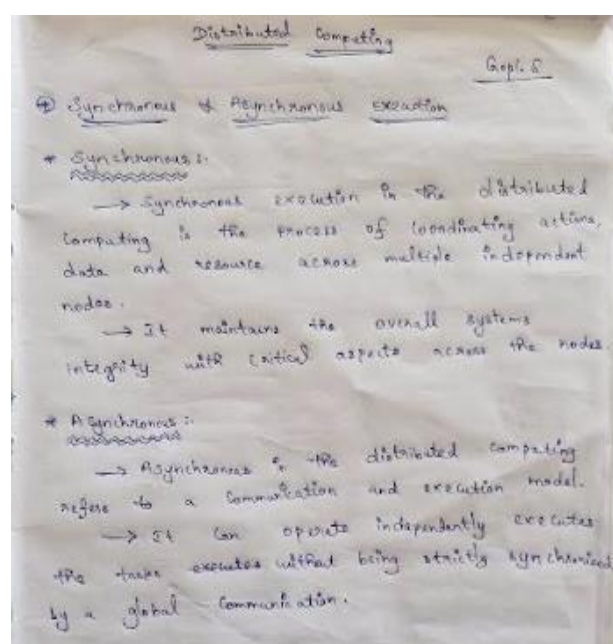
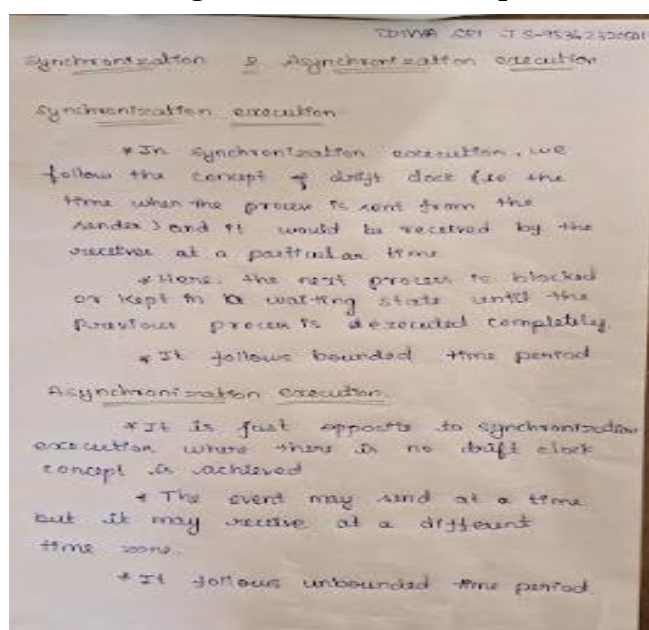


Fig 1: Sample One minute paper activity Written by Mr. Gopi S and Ms. Divya Sri J

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students mentioned that the activity helped them effectively recall techniques and concepts related to synchronous and asynchronous executions.
- They noted that this method would be helpful during exam preparation and for reinforcing core concepts in computing systems and software development.

❖ *Benefit of the practice:*

- The activity significantly improved students' retention of the topic by encouraging active recall and reflection.
- It also gave the faculty valuable insights into students level of understanding and helped identify specific learning gaps.

❖ *Challenges faced in implementation:*

- A few students initially misunderstood the concept and needed additional clarification.
- A few students took longer than expected to finish the one-minute summary task.
- Several students required additional time to complete the summary within the given timeframe.

References

- <https://www.geeksforgeeks.org/difference-between-synchronous-and-asynchronous-programming/>
- <https://www.ibm.com/cloud/learn/synchronous-vs-asynchronous>
- <https://www.freecodecamp.org/news/synchronous-vs-asynchronous-javascript-explained/>

Signature of Faculty Member

HOD